

Neel Faganiya

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Education

University of Waterloo

Master of Mathematics (M.Math.) in Computer Science (CrySP group)

Waterloo, ON

Sep. 2025 – Present

- **GPA:** 4.0/4.0
- **Advisor:** Dr. Urs Hengartner
- **Research Areas:** User Authentication, which includes authenticating users based on their behavioural biometrics.

Toronto Metropolitan University

Bachelor of Science (Honours), Computer Science (Co-op)

Toronto, ON

2020 – 2025

- GPA: 4.13/4.33

Dean's List

Research Projects

Making Sense Together: Human–AI Collaboration in Literature Review (University of Waterloo — CS 889: W26)

- Designed and evaluated an AI-assisted literature review system that improves collaborative sensemaking through transparent interaction design, steerable search, and user-centered interface features. [\[Report\]](#) [\[Code\]](#)

How do Language Models Encode Privacy Bias?

(University of Waterloo — CS 858: F25)

- Graduate research project investigating privacy bias in Large Language Models using Mechanistic Interpretability and the Contextual Integrity framework. [\[Report\]](#)

Industry Experience

Royal Bank of Canada

Cyber Security Analyst Intern

May 2024 – Aug. 2024

Toronto, ON

- Collaborated with cross-functional teams at Royal Bank of Canada and City National Bank to successfully deploy the Venafi certificate lifecycle management platform across enterprise environments.
- Designed and implemented an automated certificate provisioning workflow using Venafi and Difenda, reducing manual certificate management effort and improving operational efficiency.
- Contributed to the PKI Governance initiative by streamlining certificate onboarding and compliance processes, reducing the pending application backlog by approximately 30%.
- Worked with infrastructure, application, and security teams to troubleshoot certificate deployment issues and ensure secure PKI integration across multiple enterprise systems.

Royal Bank of Canada

Cyber Security Analyst Intern

Sep. 2022 – Aug. 2023

Toronto, ON

- Led the enterprise-wide deployment of the CA Trust Chain Package across RBC applications and servers, improving certificate trust management and organizational security compliance.
- Developed operational risk metrics and delivered technical presentations to stakeholders, supporting risk identification, reporting, and governance initiatives.
- Led weekly technical consultation sessions with application owners, providing guidance on certificate installation, renewal, and troubleshooting across Windows, Linux, and proprietary platforms.
- Partnered with infrastructure and application teams to resolve complex PKI-related issues, ensuring reliable certificate deployment throughout the organization.

Teaching Experience

University of Waterloo

CS 453/653: Software and Systems Security

Waterloo, ON

Spring 2026

CS 453/698: Software and Systems Security

Winter 2026

CS 135: Introduction to Functional Programming

Fall 2025

Selected Projects

AdVista

React, Redux, Node.js, WebSockets

- Led the development of a scalable marketplace platform for Toronto Metropolitan University students using React, Redux, and Node.js.
- Built a responsive user interface and integrated a real-time chat system using WebSockets to improve user communication and engagement.

Real-Time Multiplayer Battleship

JavaScript, Node.js, WebSockets

- Developed a real-time multiplayer Battleship game using JavaScript, Node.js, and WebSockets with synchronized gameplay and interactive audio/visual feedback.
- Designed a secure client-server communication protocol to prevent unauthorized access to game state and ensure fair gameplay.

FreeHit Cricket Database Management System

Java, JavaFX, MySQL

- Designed and implemented a normalized relational database with full CRUD functionality for cricket statistics and player management.
- Developed a JavaFX desktop application with an intuitive graphical interface for querying, updating, and managing records.

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: React, Node.js, ASP.NET, Redux, Streamlit

Security: PKI, TLS, OWASP Top 10, MITRE ATT&CK, Wireshark, Metasploit

Developer Tools: Git, Linux, Docker, MongoDB, Tableau, ServiceNow